



NORTHERN BEACHES SECONDARY COLLEGE

MANLY SELECTIVE CAMPUS

Year 12

Trial Examination

2020

Mathematics Extension II

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Write your name on the front of every booklet.
- In Questions 11 to 16 show relevant mathematical reasoning and/or calculations.
- NESA approved calculators and templates may be used.
- Weighting: 45%

Section I Multiple Choice

- 10 marks
- Attempt all questions.
- Answer Sheet provided
- Allow about 15 minutes for this section

Section II Free Response

- 90 marks
- Start a separate booklet for each question.
- Each question is of equal value.
- All necessary working should be shown in every question.
- Allow about 2 hour and 45 minutes for this section.

Section I

10 marks

Attempt Questions 1 – 10

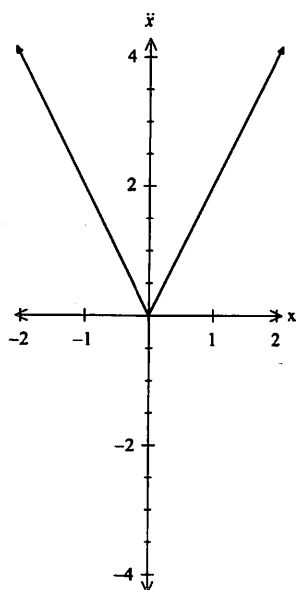
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

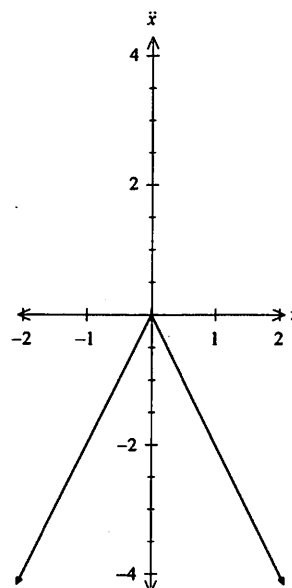
- Q1. A particle is moving along a straight line. The displacement of the particle from a fixed point O is given by x . The graphs below show acceleration $\ddot{x}$ as a function of displacement x .

Which one of the graphs below best represents a particle moving in simple harmonic motion?

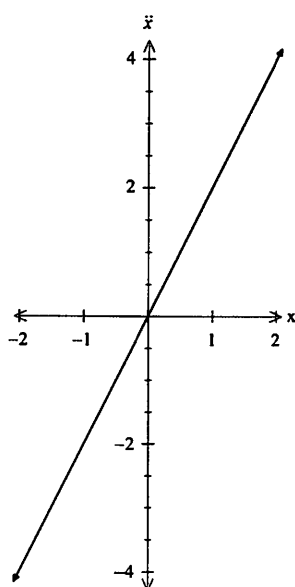
A



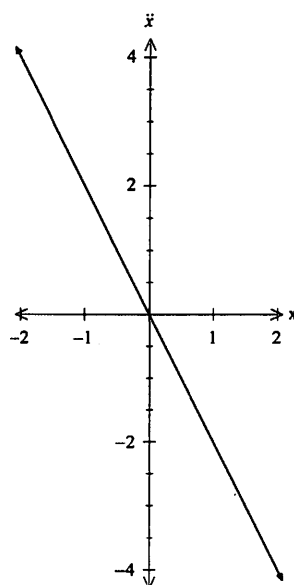
B



C



D



Q2. Which integral has the smallest value?

A. $\int_0^{\frac{\pi}{4}} \sin^2 x \, dx$

B. $\int_0^{\frac{\pi}{4}} \cos^2 x \, dx$

C. $\int_0^{\frac{\pi}{4}} \sin x \cos x \, dx$

D. $\int_0^{\frac{\pi}{4}} \sin x \tan x \, dx$

Q3. Which expression is equal to $\int \frac{1}{\sqrt{1-4x^2}} \, dx$?

A. $\frac{1}{2} \sin^{-1} \frac{x}{2} + C$

B. $\frac{1}{2} \sin^{-1} 2x + C$

C. $\sin^{-1} \frac{x}{2} + C$

D. $\sin^{-1} 2x + C$

Q4. Which symbol belongs in the box in the following statement?

$$y = e^x \quad \boxed{} \quad x = \log_e y$$

A. $\forall$

B. $\exists$

C. $\Rightarrow$

D. $\Leftrightarrow$

Q5. The following solution demonstrates that $\sqrt{2}$ is an irrational number

Let $\sqrt{2} = \frac{p}{q}$, where p and q are positive integers and have no common factors

$$2 = \frac{p^2}{q^2} \text{ on squaring both sides}$$

$$p^2 = 2q^2$$

$\therefore p^2$ is even, $\therefore p$ is even

Let $p = 2k$, where k is an integer

$$4k^2 = 2q^2$$

$$\therefore q^2 = 2k^2$$

$\therefore q^2$ is even, $\therefore q$ is even.

Which method was used in this proof?

A. Direct proof

B. Proof by contradiction

C. Proof by contrapositive

D. Proof by counter-examples.

Q6. Given that $w^5 = 1$ and w is a complex number what is the value of

$$1 + w + w^2 + w^3 + w^4 + w^5 \quad ?$$

A. 1

B. 0

C. w

D. $-w$

Q7. Let $\arg(z) = \frac{\pi}{5}$ for a certain complex number z . What is $\arg(z^7)$?

A. $-\frac{7\pi}{5}$

B. $-\frac{3\pi}{5}$

C. $\frac{2\pi}{5}$

D. $\frac{3\pi}{5}$

Q8. If $\underline{a} = -2\underline{i} - \underline{j} + 3\underline{k}$ and $\underline{b} = -m\underline{i} + \underline{j} + 2\underline{k}$ where m is a real constant, the vector $\underline{a} - \underline{b}$ will be perpendicular to the vector $\underline{b}$ when m equals

A. 0 only

B. 2 only

C. 0 or 2

D. 0 or -2

Q9. Points A , B and C have position vectors

$\underline{a} = 2\underline{i} + \underline{j}$, $\underline{b} = 3\underline{i} - \underline{j} + \underline{k}$ and $\underline{c} = -3\underline{j} + \underline{k}$ respectively.

The cosine of the angle ABC is equal to ?

A. $\frac{7}{\sqrt{6}\sqrt{13}}$

B. $-\frac{1}{\sqrt{6}\sqrt{13}}$

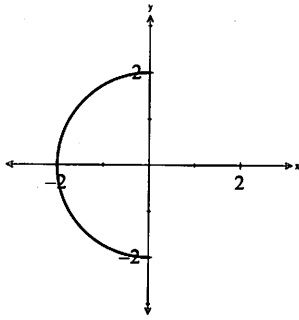
C. $-\frac{7}{\sqrt{21}\sqrt{6}}$

D. $-\frac{2}{\sqrt{6}\sqrt{13}}$

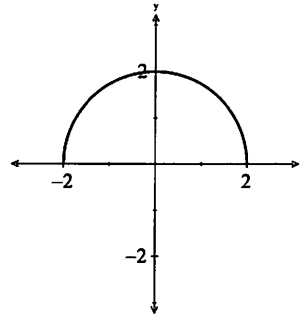
Q10. Which of the following graphs is correct for the following parametric function?

$$x = -|2\cos t| \quad y = |2\sin t|$$

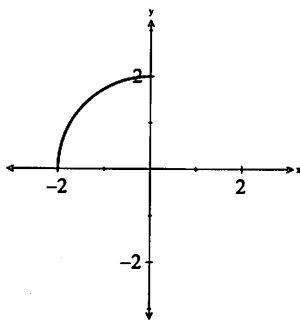
A



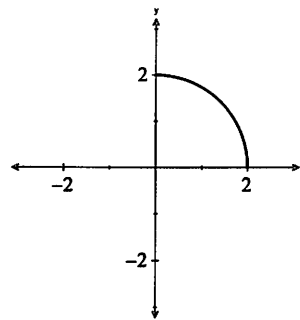
B



C



D



End of Multiple Choice

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11

15 marks

- a) Consider the following statement.

$$\forall m \in \mathbb{Z}^+ \exists n \in \mathbb{Z}^+ \text{ such that } \frac{1}{n} - \frac{1}{n+1} < \frac{1}{m}$$

- i. Write in symbolic notation the negation of the above statement
(without the negation symbol present). 1
- ii. Prove that the original statement is true. 1

- b) Use the substitution $t = \tan\left(\frac{x}{2}\right)$ to find $\int \frac{dx}{1 + \cos x - \sin x}$ 3

- c) Show by Mathematical Induction that $8^n + 2 \times 7^n - 1$ is divisible by 7 for $n \geq 1$ 3

Question 11 continues on the next page.

Question 11 continued.

d) A complex number z is defined by the equation $\arg(z - 2 - i) = \frac{3\pi}{4}$.

i. Plot on an argand diagram all the points which satisfy this relationship. **2**

ii. What is the minimum value that $|z|$ can take? **1**

e) Using the substitution $u^2 = 4 - x^2$, or otherwise, evaluate **4**

$$\int_0^2 x^3 \sqrt{4 - x^2} \, dx$$

End of Question 11

Question 12**15 marks**

- a) i. Use the method of partial fractions to show that 3

$$\int \frac{2}{x^3 + x^2 + x + 1} dx = \log_e \left| \frac{x+1}{\sqrt{x^2+1}} \right| + \tan^{-1} x + C$$

- ii. Hence show that 3

$$\int_{\frac{1}{2}}^2 \frac{2}{x^3 + x^2 + x + 1} dx = \tan^{-1} \left(\frac{3}{4} \right)$$

- b) The displacement x of a particle P moving along a straight line with respect to a fixed point O is given by

$$x = 6 \sin \left(2t + \frac{\pi}{4} \right) + \sin 2t$$

- i. Show that P is moving in simple harmonic motion about O . 2
ii. State the period of the motion. 1
iii. Find the amplitude of the motion. Leave your answer in exact form. 2

c) Let $z = 2e^{\frac{2\pi i}{3}}$ and $w = \sqrt{2}e^{\frac{i\pi}{4}}$

- i. Convert z and w to Cartesian form 1
ii. Find $\frac{z}{w}$ in Cartesian form and polar form. 2
iii. Hence, find the exact values of $\cos\left(\frac{5\pi}{12}\right)$ and $\sin\left(\frac{5\pi}{12}\right)$ 1

End of Question 12

Question 13

15 marks

- a) The point $B \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ is on the interval AC and is twice as far from $A \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ as it is from $C \begin{pmatrix} 4 \\ -1 \\ 5 \end{pmatrix}$. Use vectors to find the coordinates of B . 2

- b) A particle is moving in such a way that its speed $v \text{ ms}^{-1}$ is given by

$$v^2 = 2 - x - x^2$$

where x is the displacement of the particle from a fixed point O .

- i. Show that the particle is moving in simple harmonic motion. 2
- ii. What is the maximum distance of the particle from the origin? 2
- iii. The particle is initially at the point $x = -\frac{1}{2}$. 1

At what time does the particle first return to this point?

- c) Given that $a_1 = 2$ and $a_n = \frac{a_{n-1}}{n}$ for $n \geq 2$, prove that $a_n = \frac{2}{n!}$ for $n \geq 1$ 3

- d) Let n be a natural number.

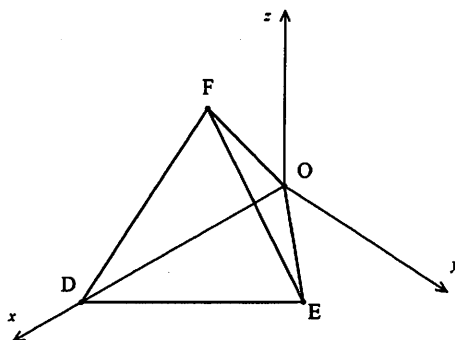
- i. Show that if n is composite, then there exists a factor of n not equal to 1, and at most equal to $\sqrt{n}$. 2
- ii. State the converse of the proposition in part i. 1
- iii. Write down the contrapositive of the proposition in part i. 2

Question 13 completed.

Question 14

15 Marks

- a) Evaluate $\int_0^1 \tan^{-1} x \, dx$. Express your answer in simplest exact form. 3
- b) i. Prove that $\frac{x}{y} + \frac{y}{x} \geq 2$ where $x, y > 0$ 2
- ii. Hence, or otherwise, show that $(x + y)\left(\frac{1}{x} + \frac{1}{y}\right) \geq 4$ 1
- iii. Deduce that $(x + y + z)\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right) \geq 9$ 2
- c) Find all the solutions for θ given $|e^{2i\theta} - 1| = \sqrt{3}$ satisfying $-\pi < \theta < \pi$ 3
- d) The faces of the tetrahedron $ODEF$ are equilateral triangles of side length 1 unit. Its base ODE lies flat on the xy plane with two vertices at O and $D(1,0,0)$ with F above the xy plane.



- i. Show that the coordinates of E are $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}, 0\right)$ 1
- ii. Using vectors, prove the coordinates of the vertex F are $\left(\frac{1}{2}, \frac{\sqrt{3}}{6}, \frac{\sqrt{6}}{3}\right)$. 3

Question 14 completed.

Question 15**15 marks**

a) The line l_1 has the equation $\vec{r} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$.

The line l_2 has the equation $\vec{r} = \begin{pmatrix} 1 \\ 3 \\ 6 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$

Show that l_1 and l_2 do not meet.

3

b) It is given that a and b are positive real numbers.

Consider the statement $\forall a (\forall b, a^{\ln b} = b^{\ln a})$.

Either prove the statement is true or provide a counter example.

2

c) Let $P(x) = x^4 + 16x^3 + 108x^2 + 400x + 800$. $P(x)$ has roots $a + 2ib$ and $3a + ib$, where $a, b \in \mathbb{R}$.

i. Find all the roots of $P(x)$ with integer values for a and b .

3

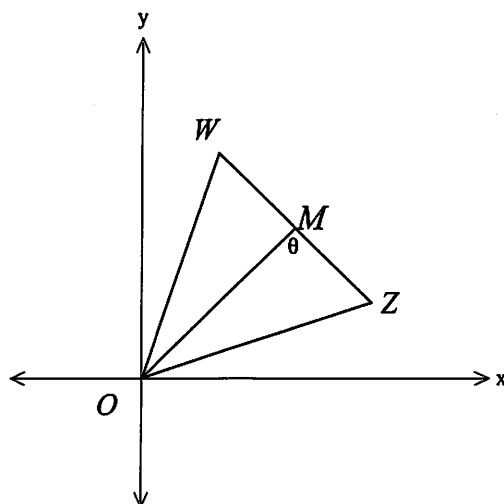
ii. Factorise $P(x)$ into its quadratic factors with real coefficients.

2

Question 15 continues on the next page.

Question 15 continued.

- d) Let the complex numbers $W(w)$, $Z(z)$ and the origin form a triangle of area $1u^2$ on the Argand diagram. M is the midpoint of WZ . Let $\angle ZMO = \theta$.



- i. Show that $|z + w| |z - w| \sin \theta = 4$ 2
- ii. Prove that $|z|^2 + |w|^2 = \frac{|z + w|^2 + |z - w|^2}{2}$ 2
- iii. If $\theta = \frac{\pi}{2}$, prove that $(|z| - |w|)(|z| + |w|) = 0$ 1

Question 15 completed

Question 16**15 marks**

- a) Write the Cartesian equation $3x - 4y = 11$ as a vector equation. 2
- b) The point $\vec{v}$ is a point on the surface of a sphere with centre $P(1,3,1)$ and radius 14 units.
- i. Write down the vector equation of the sphere. 1
- ii. The point $Q(2,1,4)$ lies on the surface of the sphere.
Find the Cartesian equation of the tangent to the sphere at Q . 3
- (You may assume that the radius drawn to the point of contact of the tangent is perpendicular to the tangent.)
- c) By considering $f'(x)$ where $f(x) = e^x - x$
- i. Show that $e^x > x$ for $x \geq 0$ 2
- ii. Hence, use a calculus method and Mathematical Induction to show that for
 $x \geq 0$, $e^x \geq \frac{x^n}{n!}$ for all $n \in \mathbb{Z}^+$ 3
- d) Find $\int \frac{\log_e x}{(1 + \log_e x)^2} dx$ 4

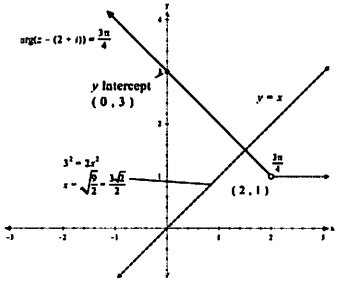
End of paper

Q1	$\ddot{x} = -n^2 x$ ie. Acceleration is a linear function opposite in sign to x	D
Q2	$0 < x < \frac{\pi}{4}, 0 < \sin x < \cos x < 1$ $\therefore \sin^2 x < \cos^2 x$ $\sin x \times \sin x < \sin x \times \cos x$ $\therefore \sin^2 x < \sin x \cos x$ $\text{because } 0 < \cos x < 1$ $\sin x < \frac{\sin x}{\cos x}$ $\sin x < \tan x$ $\sin^2 x < \sin x \tan x$ $\therefore \int_0^{\frac{\pi}{4}} \sin^2 x \, dx \text{ has the smallest value}$	A
Q3	$\int \frac{1}{\sqrt{1-4x^2}} \, dx$ $= \int \frac{1}{\sqrt{4\left(\left(\frac{1}{2}\right)^2 - x^2\right)}} \, dx$ $= \frac{1}{2} \int \frac{1}{\sqrt{\left(\frac{1}{2}\right)^2 - x^2}} \, dx$ $= \frac{1}{2} \sin^{-1}(2x) + C$	B
Q4	$\Leftrightarrow$ Implies there is an x value such that $x = \log_e y$	C
Q5	Proof by contradiction	B
Q6	$w^5 - 1 = 0$ $\Sigma \alpha = -\frac{b}{a}$ $1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$ $\therefore 1 + \omega + \omega^2 + \omega^3 + \omega^4 + \omega^5 = 1$	A

Q7	$\arg(z) = \frac{\pi}{5}$ $\therefore \arg(z^7) = \frac{7\pi}{5} = -\frac{3\pi}{5}$	B
Q8	Perpendicular $\therefore (a-b) \cdot b = 0$ $a = -2i - j + 3k$ $b = -mi + j + 2k$ $a-b = (-2i-j+3k) - (-mi+j+2k)$ $= (m-2) - 2j + k$ $(a-b) \cdot b = -m(m-2) - 2 \times 1 + 2 \times 1$ $= -m(m-2)$ $-m(m-2) = 0$ $m = 0 \text{ or } m = 2$	C
Q9	Application of the scalar product to $\overrightarrow{BA}$ and $\overrightarrow{BC}$ gives the required result. $\cos \theta = \frac{(\underline{b} - \underline{a}) \cdot (\underline{b} - \underline{c})}{ \underline{b} - \underline{a} \underline{b} - \underline{c} }$ $\underline{b} - \underline{a} = (3\underline{i} - \underline{j} + \underline{k}) - (2\underline{i} + \underline{j}) = \underline{i} - 2\underline{j} + \underline{k}$ $\underline{b} - \underline{c} = (3\underline{i} - \underline{j} + \underline{k}) - (-3\underline{j} + \underline{k}) = 3\underline{i} + 2\underline{j}$ $ \underline{b} - \underline{a} = \sqrt{6}$ $ \underline{b} - \underline{c} = \sqrt{13}$ $(\underline{b} - \underline{a}) \cdot (\underline{b} - \underline{c}) = 3 - 4 + 0 = -1$ $\cos \theta = -\frac{1}{\sqrt{6}\sqrt{13}}$	B
Q10	$x^2 + y^2 = 4 \quad x < 0, y > 0$	C

Q11		
a-i	$\forall m \in \mathbb{Z}^+ \exists n \in \mathbb{Z}^+$ such that $\frac{1}{n} - \frac{1}{n+1} < \frac{1}{m}$ Negation $\forall n \in \mathbb{Z}^+ \exists m \in \mathbb{Z}^+$ such that $\frac{1}{n} - \frac{1}{n+1} \geq \frac{1}{m}$	1 – correct answer
a-ii	$\forall m \in \mathbb{Z}^+ \exists n \in \mathbb{Z}^+$ such that $\frac{1}{n} - \frac{1}{n+1} < \frac{1}{m}$ let $m = n$ and $n = \text{any number}$ $\begin{aligned} \text{LHS} &= \frac{1}{n} - \frac{1}{n+1} \\ &= \frac{(n+1) - n}{n(n+1)} \\ &= \frac{1}{n(n+1)} \\ &< \frac{1}{n} \\ \therefore &< \frac{1}{m} \\ \therefore & \end{aligned}$ $\forall m \in \mathbb{Z}^+ \exists n \in \mathbb{Z}^+$ such that $\frac{1}{n} - \frac{1}{n+1} < \frac{1}{m}$	2 marks – correct answer demonstrating true for all m 1 mark – correct demonstration of a value of m but not all values.

b	$\int \frac{dx}{1 + \cos x - \sin x}$ $t = \tan \frac{x}{2}$ $\tan^{-1}(t) = \frac{x}{2}$ $\frac{1}{1+t^2} dt = \frac{1}{2} dx$ $\frac{2}{1+t^2} dt = dx$ $\int \frac{1}{\left(1 + \frac{1-t^2}{1+t^2} - \frac{2t}{1+t^2}\right)} \times \frac{2}{1+t^2} dt$ $= \int \frac{1+t^2}{(1+t^2+1-t^2-2t)} \times \frac{2}{1+t^2} dt$ $= \int \frac{2}{2-2t} dt$ $= \ln 1-t + C$ $= \ln\left(1 - \tan \frac{x}{2}\right) + C$	3 marks – correct solution 2 marks – forms correct integrand to $\frac{2}{2-2t}$ 1 mark – - Obtains a correct transformation of dx - Correctly integrates an incorrect expression of equivalent merit.
c	$8^n + 2 \times 7^n - 1$ Let $n=1$ $8^1 + 2 \times 7^1 - 1 = 8 + 14 - 1 = 21 = 3 \times 7$ Assume true for $n=k$ ie. $8^k + 2 \times 7^k - 1 = 7P$ where P element of $\mathbb{Z}$ $\therefore$ $8^k = 7P - 2 \times 7^k + 1$ RTP true for $n=k+1$ $8^{k+1} + 2 \times 7^{k+1} - 1 = 7M$ where M element of $\mathbb{Z}$ $\text{LHS} = 8^{k+1} + 2 \times 7^{k+1} - 1$ $= 8 \times 8^k + 2 \times 7 \times 7^k - 1$ $= 8 \times (7P - 2 \times 7^k + 1) + 2 \times 7 \times 7^k - 1$ $= 8 \times 7P - 2 \times 7^k + 7$ $= 7(8P - 2 \times 7^{k-1} + 1)$ $= 7M$ where M element of $\mathbb{Z}$ as $P \in \mathbb{Z}$ QED by Mathematical Induction	3 marks – correct solution 2 marks – correctly established $S(1)$, stated $S(k)$ and reduces exponents in $S(k+1)$, (line 2 from LHS). 1 mark - correctly established $S(1)$, correctly stated $S(k)$ and $S(k+1)$

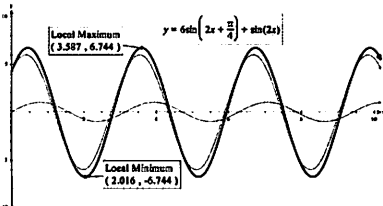
d-i	$\arg(z - 2 - i) = \frac{3\pi}{4}$ $\arg(z - (2 + i)) = \frac{3\pi}{4}$ Solution is red line with open circle and correct argument. 	2 marks – correct solution 1 mark – correct solution without open circle.
d-ii	Minimum value = perp distance $x = \frac{3\sqrt{2}}{2}$	1 mark – correct answer.

e	$\int_0^2 x^3 \sqrt{4 - x^2} \, dx$ $u^2 = 4 - x^2$ $x = 0 \Rightarrow u = 2$ $x = 2 \Rightarrow u = 0$ $2u \, du = -2x \, dx$ $u \, du = -x \, dx$ $x^2 = 4 - u^2$ $\int_0^2 x^2 \sqrt{4 - x^2} \, x \, dx$ $= - \int_0^2 x^2 \sqrt{4 - x^2} \, -x \, dx$ $= - \int_2^0 (4 - u^2) \sqrt{u^2} \, u \, du$ $= \int_0^2 4u^2 - u^4 \, du$ $= \left[\frac{4u^3}{3} - \frac{u^5}{5} \right]_0^2$ $= \frac{32}{3} - \frac{32}{5}$ $= \frac{64}{15}$	4 marks – correct solution 3 marks – correct transformation of differential, integrand and limits 2 marks – correct transformation of any two differential, limits and integrand. 1 mark – correct transformation of any <u>one</u> differential, integrand or limits.
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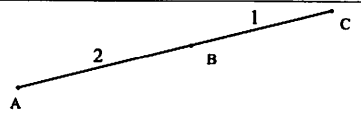
Question 12

a-i	$\int \frac{2}{x^3 + x^2 + x + 1} dx$ $\frac{2}{x^3 + x^2 + x + 1}$ $= \frac{2}{x^2(x+1) + (x+1)}$ $= \frac{2}{(x^2+1)(x+1)}$ $\frac{2}{(x^2+1)(x+1)} = \frac{Ax+B}{x^2+1} + \frac{C}{x+1}$ $(Ax+B)(x+1) + C(x^2+1) = 2$ $Ax^2 + Ax + Bx + B + Cx^2 + C = 2$ $\therefore \begin{aligned} A+C &= 0 \Rightarrow C = -A \\ A+B &= 0 \Rightarrow B = -A \\ B+C &= 2 \Rightarrow -2A = 2 \end{aligned}$ $\begin{aligned} A &= -1 \\ B &= 1 \\ C &= 1 \end{aligned}$ $\int \frac{2}{x^3 + x^2 + x + 1} dx$ $= \int \frac{-x+1}{x^2+1} + \frac{1}{x+1} dx$ $= \int -\frac{x}{x^2+1} + \frac{1}{x^2+1} + \frac{1}{x+1} dx$ $= -\frac{1}{2} \ln x^2+1 + \tan^{-1} x + \ln x+1 + C$ $= \ln \left \frac{1}{\sqrt{x^2+1}} \right + \tan^{-1} x + \ln x+1 + C$ $= \ln \left \frac{x+1}{\sqrt{x^2+1}} \right + \tan^{-1} x + C$	3 marks – correct solution 2 marks – finds correct values of A, B, C and forms a correct integrand expression of three terms. 1 mark – finds correct values of A, B, C – forms a correct integrand expression from incorrect value(s) of A, B, C
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a-ii	$\left[\ln \left \frac{x+1}{\sqrt{x^2+1}} \right + \tan^{-1} x \right]_{\frac{1}{2}}$ $= \left(\ln \left \frac{2+1}{\sqrt{2^2+1}} \right + \tan^{-1} 2 \right) - \left(\ln \left \frac{\frac{1}{2}+1}{\sqrt{\frac{1}{4}+1}} \right + \tan^{-1} \frac{1}{2} \right)$ $= \left(\ln \left \frac{3}{\sqrt{5}} \right + \tan^{-1} 2 \right) - \left(\ln \left \frac{\frac{3}{2}}{\sqrt{\frac{5}{4}}} \right + \tan^{-1} \frac{1}{2} \right)$ $= \left(\ln \left \frac{3}{\sqrt{5}} \right + \tan^{-1} 2 \right) - \left(\ln \left \frac{3}{\sqrt{5}} \right + \tan^{-1} \frac{1}{2} \right)$ $= \tan^{-1} 2 - \tan^{-1} \frac{1}{2}$ $\tan^{-1} 2 - \tan^{-1} \left(\frac{1}{2} \right)$ $\tan \left[\tan^{-1} 2 - \tan^{-1} \left(\frac{1}{2} \right) \right]$ $\frac{\tan(\tan^{-1} 2) - \tan \left(\tan^{-1} \left(\frac{1}{2} \right) \right)}{1 + \tan(\tan^{-1} 2) \cdot \tan \left(\tan^{-1} \left(\frac{1}{2} \right) \right)}$ $= \frac{2 - \frac{1}{2}}{1 + 2 \times \frac{1}{2}}$ $= \frac{\frac{3}{2}}{2} = \frac{3}{4}$ $\therefore \tan \left[\tan^{-1} 2 - \tan^{-1} \left(\frac{1}{2} \right) \right] = \tan^{-1} \left(\frac{3}{4} \right)$	3 marks – correct solution 2 marks – obtains $\left(\tan^{-1} 2 - \tan^{-1} \frac{1}{2} \right)$ 1 mark – correct substitution of limits and attempts to simplify expression in exact form. – Obtains an incorrect expression involving two inverse functions.
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b-i	$x = 6\sin\left(2t + \frac{\pi}{4}\right) + \sin 2t$ $\dot{x} = 12\cos\left(2t + \frac{\pi}{4}\right) + 2\cos 2t$ $\ddot{x} = -24\sin\left(2t + \frac{\pi}{4}\right) - 4\sin 2t$ $= -4\left[6\sin\left(2t + \frac{\pi}{4}\right) + \sin 2t\right]$ $= -(2)^2 x$	2 marks – correct solution 1 mark – obtains a correct expression for $\ddot{x}$
b-ii	$\text{Period} = \frac{2\pi}{n}$ $= \frac{2\pi}{2} = \pi$	1 mark – correct answer
b-iii	 $x = 6\sin\left(2t + \frac{\pi}{4}\right) + \sin 2t$ $= 6\sin 2t \cos \frac{\pi}{4} + 6\cos 2t \sin \frac{\pi}{4} + \sin 2t$ $= \frac{6}{\sqrt{2}}\sin 2t + \frac{6}{\sqrt{2}}\cos 2t + \sin 2t$ $= (1 + 3\sqrt{2})\sin 2t + 3\sqrt{2}\cos 2t$ From auxillary angles $R = \sqrt{(1 + 3\sqrt{2})^2 + (3\sqrt{2})^2}$ $= \sqrt{1 + 6\sqrt{2} + 18 + 18}$ $= \sqrt{37 + 6\sqrt{2}}$ $\cong 6.744$	2 marks – correct solution 1 mark - Obtains a correct expression for x in terms of $\sin 2t$ and $\cos 2t$ - Correctly evaluates R from an incorrect expression. For x.

c-i	$z = 2e^{\frac{2\pi i}{3}} \quad w = \sqrt{2}e^{\frac{i\pi}{4}}$ $re^{i\theta} = rcis\frac{\theta}{2}$ $z = 2cis\frac{2\pi}{3} \quad w = \sqrt{2}cis\frac{\pi}{4}$ $= 2\left\{-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right\} \quad w = \sqrt{2}\left\{\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i\right\}$	1 mark – both correct.
c-ii	$\frac{z}{w} = \frac{ z }{ w }cis[\arg z - \arg w]$ $= \frac{2}{\sqrt{2}}cis\left(\frac{2\pi}{3} - \frac{\pi}{4}\right)$ $= \sqrt{2}cis\left(\frac{5\pi}{12}\right)$ $= \sqrt{2}e^{\frac{5\pi i}{12}}$ $\frac{z}{w} = \frac{-1 + \sqrt{3}i}{1 + i}$ $= \frac{-1 + \sqrt{3}i}{1 + i} \times \frac{1 - i}{1 - i}$ $= \frac{(-1 + \sqrt{3}i)(1 - i)}{1^2 - i^2}$ $= \frac{-1 + i + \sqrt{3}i + \sqrt{3}}{2}$ $= \frac{-1 + \sqrt{3}}{2} + \frac{1 + \sqrt{3}}{2}i$	2 mark – correct solution 1 mark – one correct expression
c-iii	$\cos \frac{5\pi}{12} = \frac{\sqrt{3} - 1}{2}$ $\sin \frac{5\pi}{12} = \frac{\sqrt{3} + 1}{2}$	1 mark – both expressions correct.

Q13		
a-i	 Method 1 $B = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ $\overrightarrow{AB} = \frac{2}{3} \overrightarrow{AC}$ $\begin{pmatrix} 1-x \\ 2-y \\ 2-z \end{pmatrix} = \frac{2}{3} \begin{pmatrix} 1-4 \\ 2+1 \\ 2-5 \end{pmatrix}$ $1-x = \frac{2}{3} \times -3 = -2$ $x = 3$ $2-y = \frac{2}{3} \times 3 = 2$ $y = 0$ $2-z = \frac{2}{3} \times -3 = -2$ $z = 4$ $B = \begin{pmatrix} 3 \\ 0 \\ 4 \end{pmatrix}$ Method 2 $\frac{AB}{BC} = \frac{2}{1}$ $\begin{pmatrix} x-1 \\ y-2 \\ z-2 \end{pmatrix} = 2 \begin{pmatrix} 4-x \\ -1-y \\ 5-z \end{pmatrix}$ $x-1 = 8-2x \Rightarrow x = 3$ $y-2 = -2-2y \Rightarrow y = 0$ $z-2 = 10-2z \Rightarrow z = 4$	2 marks – correct solution 1 mark – finds <u>two</u> correct values for x, y, z.

b-i	$v^2 = 2 - x - x^2$ $\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$ $= \frac{1}{2} \times \frac{d}{dx} (2 - x - x^2)$ $= \frac{1}{2} (-2x - 1)$ $= -x - \frac{1}{2}$ $= -1 \times \left(x - \left(-\frac{1}{2} \right) \right)$ of form $\ddot{x} = -n^2 (x - b)$ $\therefore$ simple harmonic motion.	2 marks – correct solution 1 mark – obtains $\ddot{x} = \frac{1}{2} (-2x - 1)$
b-ii	$v^2 = 2 - x - x^2$ $v^2 \geq 0$ $2 - x - x^2 \geq 0$ $(2+x)(1-x) \geq 0$ $-2 \leq x \leq 1$ $\therefore$ maximum distance from the origin = 2 metres	2 marks – correct solution 1 mark – forms $2 - x - x^2 \geq 0$ and attempts to solve inequality.
b-iii	Initially at centre of motion $x = -\frac{1}{2}$ $n = 1, \text{ Period} = \frac{2\pi}{n} = 2\pi$ time taken to travel from centre and return to centre = $\frac{1}{2} \times \text{period}$ $\therefore t = \pi$ seconds	1 mark – correct answer

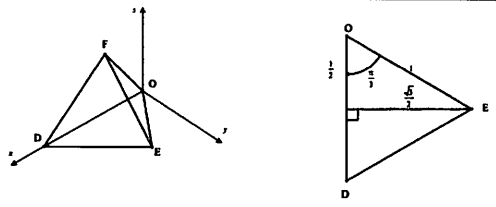
d-ii	$a_1 = 2, a_n = \frac{a_{n-1}}{n} \text{ for } n \geq 2 \Rightarrow a_n = \frac{2}{n!} \text{ for } n \geq 1$ $a_1 = \frac{2}{1!} = 2$ $a_2 = \frac{a_1}{2} = \frac{2}{2} = \frac{2}{2!}$ Assume true for $n = k$ ie. $\Rightarrow a_k = \frac{2}{k!}$ RTP for $n = k+1$ ie. $\Rightarrow a_{k+1} = \frac{2}{(k+1)!}$ $a_{k+1} = \frac{a_k}{k+1}$ $= \frac{2}{k!} \times \frac{1}{(k+1)}$ $= \frac{2}{(k+1)k!}$ $= \frac{2}{(k+1)!}$ QED by process of Mathematical induction.	3 marks – correct solution. 2 marks- correct solution for S(1), correct statement for S(k) and S(k+1) and attempts to show S(k)$\Rightarrow$S(k+1) 1 mark -correct solution for S(1), correct statement for S(k)								
d-i	If composite then Let $n = pq$ where $p < n, q < n$ for $n > 1$ If $p \leq \sqrt{n}$ then proposition is true If $p > \sqrt{n}$ then $pq = n \Rightarrow q < \sqrt{n}$ $\therefore$ proposition is true	2 marks – correct solution. 1 mark – one correct statement for $p \leq \sqrt{n}$ or $p > \sqrt{n}$								
d-ii	<table> <tr> <td>Initial proposition</td> <td>Converse</td> </tr> <tr> <td>$P = n$ is a composite number</td> <td>n has a factor less than $\sqrt{n}$</td> </tr> <tr> <td>$Q =$ a factor $\leq \sqrt{n}$</td> <td>$\Rightarrow n$ is composite</td> </tr> <tr> <td>$P \Rightarrow Q$</td> <td></td> </tr> </table>	Initial proposition	Converse	$P = n$ is a composite number	n has a factor less than $\sqrt{n}$	$Q =$ a factor $\leq \sqrt{n}$	$\Rightarrow n$ is composite	$P \Rightarrow Q$		1 mark – correct answer
Initial proposition	Converse									
$P = n$ is a composite number	n has a factor less than $\sqrt{n}$									
$Q =$ a factor $\leq \sqrt{n}$	$\Rightarrow n$ is composite									
$P \Rightarrow Q$										

d-iii	Contrapositive: $\neg Q \Rightarrow \neg P$ If n does not have factor $\leq \sqrt{n}$ then not composite (ie prime) If A then B If n is composite, then there exists a factor of n not equal to 1 or itself, and at most equal to $\sqrt{n}$. Contrapositive If NOT B then NOT A If a factor exists greater than $\sqrt{n}$ then n is NOT composite.	2 marks – correct solution 1 mark – attempts to formulate a statement 7Q$\Rightarrow$
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Question 14

a	$I = \int_0^1 \tan^{-1} x \, dx$ $= \int_0^1 1 \times \tan^{-1} x \, dx$ $u = \tan^{-1} x \quad dv = 1$ $du = \frac{1}{1+x^2} \quad v = x$ $I = \left[x \tan^{-1} x \right]_0^1 - \int_0^1 \frac{x}{1+x^2} \, dx$ $= \left[x \tan^{-1} x \right]_0^1 - \frac{1}{2} \left[\ln 1+x^2 \right]_0^1$ $= [\tan^{-1} 1 - 0] - 1.2[\ln 2 - \ln 1]$ $= \frac{\pi}{4} - \ln(\sqrt{2})$	3 marks – correct solution 2 marks – finds a correct primitive function for $\int \tan^{-1} x \, dx$ 1 mark – finds correct expressions for du and dv.
b-i	$\frac{x}{y} + \frac{y}{x} \geq 2$ $(x-y)^2 \geq 0$ $x^2 - 2xy + y^2 \geq 0$ $x^2 + y^2 \geq 2xy$ $x \Rightarrow \sqrt{\frac{x}{y}}$ $y \Rightarrow \sqrt{\frac{y}{x}}$ $\left(\sqrt{\frac{x}{y}} \right)^2 + \left(\sqrt{\frac{y}{x}} \right)^2 \geq 2 \sqrt{\frac{x}{y}} \sqrt{\frac{y}{x}}$ $\left(\sqrt{\frac{x}{y}} \right)^2 + \left(\sqrt{\frac{y}{x}} \right)^2 \geq 2 \sqrt{\frac{x}{y} \times \frac{y}{x}}$ $\left(\sqrt{\frac{x}{y}} \right)^2 + \left(\sqrt{\frac{y}{x}} \right)^2 \geq 2$ Or any similar method.	2 marks – correct solution 1 mark – states or derives $x^2 + y^2 \geq 2xy$ and identifies $x \Rightarrow \sqrt{\frac{x}{y}}$ and/or $y \Rightarrow \sqrt{\frac{y}{x}}$

	Hence method $(x+y)\left(\frac{1}{x} + \frac{1}{y}\right)$ $= 1 + \frac{x}{y} + \frac{y}{x} + 1$ $= 2 + \frac{x}{y} + \frac{y}{x}$ $\geq 2 + 2$ ≥ 4 Alternate method $x + y \geq 2\sqrt{xy}$ $(x+y)^2 \geq 4xy$ $\frac{(x+y)^2}{xy} \geq 4$ $\frac{1}{x} + \frac{1}{y} = \frac{y+x}{xy}$ $\therefore (x+y) \frac{y+x}{xy}$ $= \frac{(x+y)^2}{xy}$ $\therefore (x+y)\left(\frac{1}{x} + \frac{1}{y}\right) \geq 4$	1 mark – a correct solution.
b-ii		
b-iii	$(x+y+z)\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right)$ $= (x+y)\left(\frac{1}{x} + \frac{1}{y}\right) + (x+y)\frac{1}{z} + z\left(\frac{1}{x} + \frac{1}{y}\right) + z \times \frac{1}{z}$ $\geq 4 + 1 + (x+y)\frac{1}{z} + z\left(\frac{1}{x} + \frac{1}{y}\right)$ $\geq 5 + \left(\frac{x}{z} + \frac{y}{z}\right) + \left(\frac{z}{x} + \frac{z}{y}\right)$ $\geq 5 + \left(\frac{x}{z} + \frac{z}{x}\right) + \left(\frac{y}{z} + \frac{z}{y}\right)$ $\geq 5 + 2 + 2$ ≥ 9	2 marks – correct solution 1 mark – expands expression and attempts to simplify using part (i)

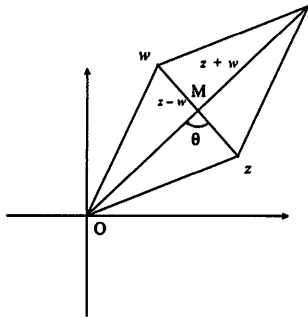
c	$ e^{2i\theta} - 1 = \sqrt{3} \quad \text{for } -\pi < \theta < \pi$ $ \operatorname{cis}(2\theta) - 1 = \sqrt{3}$ $(\cos 2\theta - 1)^2 + \sin^2 2\theta = 3$ $\cos^2 2\theta - 2\cos 2\theta + 1 + \sin^2 2\theta = 3$ $-2\cos 2\theta = 1 \quad \text{for } -2\pi < 2\theta < 2\pi$ $\cos 2\theta = -\frac{1}{2}$ $2\theta = \pm \frac{2\pi}{3}, \pm \frac{4\pi}{3}$ $\theta = \pm \frac{\pi}{3}, \pm \frac{2\pi}{3}$	3 marks - Four correct solutions 2 marks - Determines $\cos 2\theta = -\frac{1}{2}$ 1 mark - Conversion to mod/arg form
d-i	 Therefore using exact triangle ratios $E = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}, 0\right)$	1 mark – acceptable demonstration
d-ii	Method 1 In ΔFOD $\angle FOD = \angle FOE = \frac{\pi}{3}$ $\cos \angle FOD = \frac{(OF) \cdot (OD)}{ OF OD }$ $\cos \frac{\pi}{3} = \frac{(x,y,z) \cdot (1,0,0)}{1 \times 1}$ $\frac{1}{2} = x$ In ΔFOE $\cos \angle FOE = \frac{(x,y,z) \cdot \left(\frac{1}{2}, \frac{\sqrt{3}}{2}, 0\right)}{1 \times 1}$ $\frac{1}{2} = \frac{x}{2} + \frac{\sqrt{3}y}{2} + 0$ $x + \sqrt{3}y = 1$ $\frac{1}{2} + y\sqrt{3} = 1$ $y = \frac{1}{2\sqrt{3}} = \frac{\sqrt{3}}{6}$	3 marks – x, y, z correctly derived 2 marks 2 of x, y, z correctly derived 1 mark One of x, y, z correctly derived

	$ \vec{OF} = \sqrt{x^2 + y^2 + z^2}$ $1 = \sqrt{\frac{1}{4} + \frac{1}{12} + c^2}$ $c^2 = 1 - \frac{1}{4} - \frac{1}{12} = \frac{2}{3}$ $c = \sqrt{\frac{2}{3}} = \frac{\sqrt{6}}{3}$ $F = \left(\frac{1}{2}, \frac{\sqrt{3}}{5}, \frac{\sqrt{6}}{3} \right)$	
d-ii	Alternate acceptable vector approach.	

Q15		
a	$l_1 : r = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \quad l_2 : r = \begin{pmatrix} 1 \\ 3 \\ 6 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$ Comparing k components $-1 = 6 - \mu$ $\mu = 7$ Comparing j components $\lambda = 3 + \mu$ $\lambda = 10$ as $\mu = 7$ Comparing i components $1 + \lambda = 1 + 2\mu$ $\lambda = 2\mu$ $10 \neq 14$ Therefore, vectors do not intersect.	3 marks – correct solution 2 marks – - Using μ to find a value for λ 1 mark - Finding value for μ - Forming simultaneous equations but not solving
b	$\forall a(\forall b, a^{\log_b b} = b^{\log_b a})$ Let $a^{\log_b b} = m$ $\log_a (a^{\log_b b}) = \log_a m$ $\log_a b \log_b a = \log_a [a^{\log_b b}]$ $\log_a b \log_b a = \log_a b \log_b a$ $\log_a b \log_b a = \log_a a \log_a b$ $\log_a (a^{\log_b b}) = \log_a (b^{\log_b a})$ $(a^{\log_b b}) = (b^{\log_b a})$	2 marks – correct solution 1 mark – takes logs of both sides and completes some simplification of both sides.

	$P(x) = x^4 + 16x^3 + 108x^2 + 400x + 800$ Real coefficients therefore roots in integer pairs Therefore roots are in conjugate pairs Roots are $(a + bi), (a - bi), (3a + bi), (3a - bi)$ $\Sigma \alpha = -\frac{b}{a} = -16$ $\Sigma \alpha \beta = \frac{c}{a} = 108$ $\Sigma \alpha \beta \gamma = -\frac{d}{a} = -400$ $\Sigma \alpha \beta \gamma \theta = \frac{e}{a} = 800$ $(a + 2bi) + (a - 2bi) + (3a + bi) + (3a - bi) = -16$ $8a = -16$ $a = -2$	
c-i	$(a + 2bi) \times (a - 2bi) \times (3a + bi) \times (3a - bi) = 800$ $(a^2 + 4b^2)(9a^2 + b^2) = 800$ $(4 + 4b^2)(36 + b^2) = 800$ $144 + 148b^2 + 4b^4 = 800$ $36 + 37b^2 + b^4 = 200$ let $u = b^2$ $u^2 + 37u - 164 = 0$ $u = \frac{-37 \pm \sqrt{37^2 + 4 \times (-164)}}{2}$ $b^2 = \frac{-37 \pm 45}{2}$ $b^2 = 4 \text{ or } -82$ as b is real $b^2 = 4$ $b = \pm 2$ $\therefore$ roots are $(2 - 4i), (2 + 4i), (-6 + 2i), (-6 - 2i)$	3 marks – correct solution 2 marks – finds $a = -2$ and forms a second equation to find b 1 mark – Finds $a = -2$ or $b = -2$ or $b = 2$

c-ii	$P(x) = (x - z)(x - \bar{z}) = x^2 - 2\operatorname{Re}(z) + z ^2$ $\operatorname{Re}(-2 \pm 4i) = -2 \quad (-2 \pm 4i) ^2 = 20$ $\operatorname{Re}(-6 \pm 2i) = -6 \quad (-6 \pm 2i) ^2 = 40$ $P(x) = (x^2 - 2(-2)x + 20)(x^2 - 2(-6)x + 40)$ $= (x^2 + 4x + 20)(x^2 + 12x + 40)$	2 marks – correct solution 1 mark – Forms one correct quadratic factor – Correctly forms two correct quadratic factors from incorrect roots.
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d-i	 Area = $\frac{1}{2} ab \sin \theta$ $\Delta Owz = 1$ $\Delta Owz = \frac{1}{2} [ab \sin \theta + ab \sin(180 - \theta)]$ $= \frac{1}{2} [2ab \sin \theta]$ $1 = \frac{1}{2} z + w \frac{1}{2} z - w \sin \theta$ $4 = z + w z - w \sin \theta$	2 marks – correct solution 1 mark - Identifies $\overrightarrow{OM} = \frac{1}{2}(\underline{z} + \underline{w})$ and $\overrightarrow{MZ} = \frac{1}{2}(\underline{z} - \underline{w})$ - Or equivalent.
d-ii	Method 1 $c^2 = a^2 + b^2 - 2ab \cos \theta$ $z ^2 = \left \frac{1}{2}(z - w) \right ^2 + \left \frac{1}{2}(z + w) \right ^2 - 2 \left \frac{1}{2}(z - w) \right \left \frac{1}{2}(z + w) \right \cos \theta \dots (a)$ $w ^2 = \left \frac{1}{2}(z - w) \right ^2 + \left \frac{1}{2}(z + w) \right ^2 - 2 \left \frac{1}{2}(z - w) \right \left \frac{1}{2}(z + w) \right \cos(180 - \theta) \dots (b)$ $-\cos \theta = \cos(180 - \theta)$ $(a) + (b)$ $z ^2 + w ^2 = 2 \times \frac{1}{4} [z - w ^2 + z + w ^2]$ $z ^2 + w ^2 = \frac{ z - w ^2 + z + w ^2}{2}$	2 marks- correct solution 1 mark – forms a correct cosine rule expression for $z ^2$ or $w ^2$

d-ii	Method 2 – Dot Product method. (Shorter) $\underline{z} + \underline{w} ^2 + \underline{z} - \underline{w} ^2$ $= (\underline{z} + \underline{w}) \cdot (\underline{z} + \underline{w}) + (\underline{z} - \underline{w}) \cdot (\underline{z} + \underline{w})$ $= \underline{z} \cdot \underline{z} + 2 \underline{z} \cdot \underline{w} + \underline{w} \cdot \underline{w} + \underline{z} \cdot \underline{z} - 2 \underline{z} \cdot \underline{w} + \underline{w} \cdot \underline{w}$ $= 2 \underline{z} ^2 + 2 \underline{w} ^2$ $\therefore$ $\underline{z} ^2 + \underline{w} ^2 = \frac{ \underline{z} + \underline{w} ^2 + \underline{z} - \underline{w} ^2}{2}$	2 marks – correct solution 1 mark – writes $\underline{z} + \underline{w} ^2$ and $\underline{z} - \underline{w} ^2$ in dot product form.
d-iii	Option 1 If $\theta = \frac{\pi}{2}$ then the parallelogram must be a rhombus therefore $z = w$ $\therefore z - w = 0$ $\therefore (z - w)(z + w) = 0$	1 mark – correct answer.
d-iii	Option 2 If $\theta = \frac{\pi}{2}$ then $(\underline{z} - \underline{w}) \cdot (\underline{z} + \underline{w}) = 0$ $(\underline{z} - \underline{w}) \cdot (\underline{z} + \underline{w})$ $= \underline{z} \cdot \underline{z} + \underline{z} \cdot \underline{w} - \underline{z} \cdot \underline{w} - \underline{w} \cdot \underline{w}$ $= \underline{z} ^2 - \underline{w} ^2$ $= (\underline{z} + \underline{w})(\underline{z} - \underline{w})$ $\therefore$ $(\underline{z} + \underline{w})(\underline{z} - \underline{w}) = 0$	
	Note – do NOT use binomial expansion incorrectly ie. $(\underline{z} + \underline{w})(\underline{z} - \underline{w}) \neq \underline{z} ^2 - \underline{z} \underline{w} + \underline{z} \underline{w} - \underline{w} ^2$	

Question 16

a	$3x - 4y = 11$ $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ lies on the line Gradient of the line $\frac{3}{4}$ Vector equation $\vec{r} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 3 \end{pmatrix}$	2 marks - Correct gradient vector - A correct initial vector
b-i	Vector equation $\left \vec{r} - \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} \right = 14$ $\left \begin{pmatrix} x \\ y \\ z \end{pmatrix} - \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} \right = 14$ (Cartesian equation) $(x - 1)^2 + (y - 3)^2 + (z - 1)^2 = 14^2$	1 mark for a vector equation
b-ii	As radius is perpendicular to tangent then Radius vector = $(2 - 1, 1 - 3, (4 - 1))$ Tangent Vector = $(x - 2, y - 1, (z - 4))$ $(x - 2, y - 1, (z - 4)) \cdot (2 - 1, 1 - 3, (4 - 1)) = 0$ $(x - 2) - 2(y - 1) + 3(z - 4) = 0$ $x - 2y + 3z - 2 + 2 - 12 = 0$ $x - 2y + 3z = 12$	3 marks – correct solution 2 marks – correctly states radius and tangent vectors - Finds and equation from one/ two incorrect equations using dot product 1 mark - Finds one correct radius or tangent - Attempts to apply dot product to two vectors.

c	$f(x) = e^x - x$ let $x = 0$ $f(0) = e^0 - 0 = 1 > 0$ $f'(x) = e^x - 1 > 0$ for $x \geq 0$ as $e^x > 1$ for $x \geq 0$ $\therefore$ as $f(0) = 1 > 0$ and $f'(x) > 0$ $\therefore$ $e^x - x > 0 ; x \geq 0$ $e^x > x, x \geq 0$	2 marks – correct solution 1 mark – attempts to apply $f'(x) > 0$
	Let P_n represent the proposition $e^x \geq \frac{x^n}{n!}$ $n = 1$ – true – proven in part i ie. $e^x > x \Rightarrow e^x > \frac{x^1}{1!}$ Assume true for $n = k$ $e^x > \frac{x^k}{k!}$ RTP true for $n = k + 1$ Form assumption $e^x > \frac{x^k}{k!}$ $\int_0^x e^x dx > \int_0^x \frac{x^k}{k!} dx$ $\left[e^x \right]_0^x > \left[\frac{x^{k+1}}{(k+1)k!} \right]_0^x$ $e^x > \frac{x^{k+1}}{(k+1)!} - 0$ $e^x > \frac{x^{k+1}}{(k+1)!}$ $P(k) \Rightarrow P(k+1)$ QED by process of Mathematical Induction.	3 marks – correct solution 2 marks - show $S(1)$ is true and states $S(k)$, $S(k+1)$ correctly and attempts to use calculus method to establish the result. 1 mark – establishes $S(1)$ and correctly state $S(k)$ and $S(k+1)$

Method 1. A partial fractions approach and some observation.

$$\int \frac{\log_e x}{(1 + \log_e x)^2} dx$$

$$\int \frac{1 + \log_e x - 1}{(1 + \log_e x)^2} dx$$

$$\int \frac{1 + \log_e x}{(1 + \log_e x)^2} - \frac{1}{(1 + \log_e x)^2} dx$$

$$\int \frac{1}{(1 + \log_e x)} - \frac{1}{(1 + \log_e x)^2} dx$$

if $y = \frac{x}{1 + \ln x} = x(1 + \ln x)^{-1}$ (by observation)

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{1 + \ln x} + x \times \left(-\frac{1}{x}\right) \times \frac{1}{(1 + \ln x)^2} \\ &= \frac{1}{(1 + \log_e x)} - \frac{1}{(1 + \log_e x)^2} \end{aligned}$$

$$\therefore \int \frac{\log_e x}{(1 + \log_e x)^2} dx = \frac{x}{1 + \ln x}$$

4 marks

d Method 2 – a more traditional approach $I = \int \frac{\ln x}{(1 + \ln x)^2} dx$ let $u = \ln x \Rightarrow x = e^u$ $du = \frac{1}{x} dx \Rightarrow x du = dx$ $I = \int \frac{u}{(1 + u)^2} x du$ $= \int \frac{u}{(1 + u)^2} e^u du$ $= \int \frac{e^u \times u}{(1 + u)^2} du$ $= \int \frac{e^u \times u + e^u - e^u}{(1 + u)^2} du$ $= \int \frac{(1 + u)e^u - e^u}{(1 + u)^2} du$ $= \int \frac{(1 + u)e^u}{(1 + u)^2} du - \int \frac{e^u}{(1 + u)^2} du$ $= \int \frac{e^u}{1 + u} du - \int \frac{e^u}{(1 + u)^2} du$	4 marks- correct solution. 3 marks- $I = \int \frac{e^u}{1 + u} du - \int \frac{e^u}{(1 + u)^2} du$ Or equivalent expression, and then attempts to use IBP for $\int \frac{e^u}{(1 + u)^2} du$ 2 marks Obtains $I = \int \frac{e^u}{1 + u} du - \int \frac{e^u}{(1 + u)^2} du$ But no further progress. 1 mark – obtains $I = \int \frac{ue^u}{(1 + u)^2} du$
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$$I = \int \frac{e^u}{1+u} du - \int \frac{e^u}{(1+u)^2} du$$

Complete second integral by parts

$$I_2 = \int \frac{e^u}{(1+u)^2} du$$

$$u = e^u \quad dv = (1+u)^{-2}$$

$$du = e^u \quad v = -\frac{1}{1+u}$$

$$I_2 = \frac{-e^u}{1+u} - \int \frac{-e^u}{1+u} du$$

$$I = \int \frac{e^u}{1+u} du - I_2$$

$$= \int \frac{e^u}{1+u} du - \left\{ \frac{-e^u}{1+u} - \int \frac{-e^u}{1+u} du \right\}$$

$$= \frac{e^u}{1+u}$$

$$= \frac{e^{\ln x}}{1 + \ln x}$$

$$= \frac{x}{1 + \ln x} + C$$

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